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/ › MATLAB Drive ›

Editor

untitled8.m ×

/MATLAB Drive/untitled8.m

```
clear;
clc;
close all;
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
hexArea = @(R) (3*sqrt(3)/2) .* R.^2;

cellAreaKm2 = hexArea(radiiKm);
cellCount = ceil(serviceAreaKm2 ./ cellAreaKm2);
estimatedUsers = cellCount .* usersPerCell;
capacityTable = table( ...
    radiiKm(:), cellAreaKm2(:), cellCount(:), ...
    repmat(usersPerCell, numel(radiiKm), 1), ...
    estimatedUsers(:), ...
    'VariableNames', {'Radius_km', 'CellArea_km2', ...
    'EstimatedCells', 'UsersPerCell', 'IllustrativeUsers'});

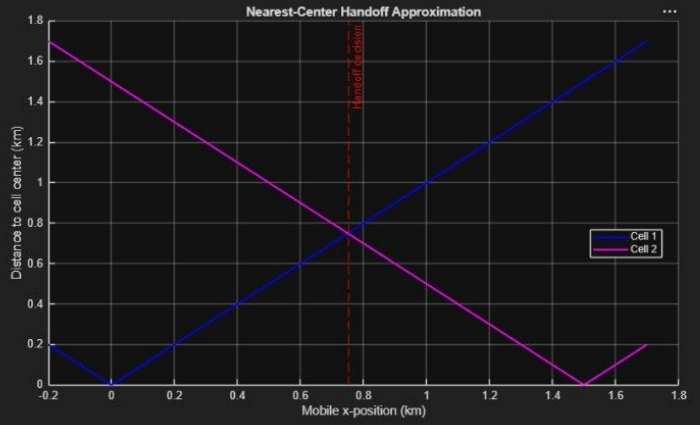
disp(capacityTable);

R = radiiKm(1);

centers = [
     0,              0;
     1.5*R,          sqrt(3)*R/2;
     1.5*R,         -sqrt(3)*R/2;
     0,              sqrt(3)*R;
     0,             -sqrt(3)*R;
    -1.5*R,          sqrt(3)*R/2;
```

Figures

Figure 1 × Figure 2 × Figure 3 ×



Command Window

New to MATLAB? See resources for Getting Started.

Radius_km	CellArea_km2	EstimatedCells	UsersPerCell	IllustrativeUsers
1	2.5981	2	30	60
0.5	0.64952	8	30	240

Geometric handoff at x = 0.753 km

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